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‘Flatten the curve’: rhetorical data visualizations of a global pandemic

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ABSTRACT

Actionable metrics within data dashboards can encourage actions to mitigate spread of COVID-19 by tying the actions to holistic data visualizations, showing a connection between an individual’s behavior and the potential impact on the overall outbreak. The analysis shows that actionable metrics are only persuasive when they take advantage of the inherent tension of holistic-localist perspectives while leveraging modes of power and visual conventions.

KEYWORDS

Actionable metrics; COVID-19; data visualization; flatten the curve (FLC); modes of power; pandemic; visual rhetoric

The call to action, “flatten the curve” (FLC), is a memorable refrain of the early years of the COVID-19 pandemic. The premise is this: if we interact with fewer people, potentially infected individuals will infect fewer people, making the virus spread much more slowly through the population. If the virus spreads at a slower rate, hospitals and other critical care facilities are less likely to be overwhelmed, the workforce will experience less of a decrease, and demand for supplies will stay low. Early in the COVID-19 pandemic, calls to FLC were usually accompanied by a simple, hypothetical disease model with the purpose of explaining the basic reproduction number, or R_0 (“r naught”). The basic reproduction number is the number of individuals infected by one infected individual. If the rate is less than one, the number of cases will decrease over time, eventually dying out. If the rate is equal to one, the number of cases will remain stable. If the rate is higher than one, infections will grow exponentially. All these scenarios assume all other contributing factors of endemicity remain the same (Sahni et al., 2020). Widely circulated visualizations, like that reproduced at IEEE ProComm (“Flatten the Curve: Why certain messages catch on,” 2020), compare two simple models: one curve that appears visually “flatter,” remaining below a horizontal “healthcare system capacity” line, and a second, steeper curve that quickly surpasses the “healthcare system capacity” threshold. The former line reflects a basic reproduction number closer to one, the latter, steeper curve reflects a larger reproduction number.

By the second year of the COVID-19 pandemic, many took a more calculated approach to social distancing and mask wearing (Stone, 2022). Although our opportunity to FLC has well passed, if we ever even had one, FLC remains an important motive for promoting public health: if you are infected (with anything), limit your exposure to others in ways that might also infect them. What we are left with is a day-to-day decision-making based on best available data. Several respected public health organizations, labs, and hospitals stepped up to the task and were quick to create online data visualizations of location-based case, death, hospitalization, and, later, vaccination rates. Technical communication scholars were equally quick to offer critiques of COVID-19 data visualizations (Amidon et al., 2021; Atherton, 2021; Carlson and Gouge, 2021; Doan, 2021; Lindgren, 2021) and data dashboards (Ivanković et al., 2021; Verhulsdonck & Shah, 2021, 2022).

Prior to COVID-19, research on disease data visualizations was relatively limited. Most notably, Candice Welhausen has analyzed early yellow fever maps (2015a), contemporary disease maps of the Ebola outbreak (2015b), and user-contributed flu maps (2017). Rather presciently, she (2015a) argued that it would be increasingly important to understand how maps construct our understanding of disease from a global health perspective. The ongoing COVID-19 pandemic certainly provides ample fodder for technical communication scholars to study how maps shape our understanding of an outbreak. Many early criticisms of COVID-19 data visualizations have analyzed a small number of static graphics, highlighting the misleading nature of specific graphics and the lack of sufficient context and its consequences (Doan, 2021; Lindgren, 2021) especially for multiply marginalized communities (Atherton, 2021; Carlson and Gouge, 2021). Compounding these findings is the more general criticisms of the overarching (FTC) metaphor that it conflates a high-level abstraction with real data driven by local, social dynamics (Amidon et al., 2021).

The value of the FLC metaphor is that it has served as a powerful call-to-action, shifting the primary motive from the individual risk of infection to the risk of infecting others. Verhulsdonck and Shah (2021) argue that “actionable metrics,” borrowed from lean methodology, can encourage actions to mitigate spread of COVID-19 by tying the actions to data visualizations. Ivanković et al. (2021), in response to Verhulsdonck and Shah (2021), argue that we must have a broader understanding of actionable metrics. Actionable metrics, according to Ivanković et al. (2021), are not merely “effective visualization and contextualized data” but must fulfill a specific information need for a specific audience at a specific time and in an understandable form. Essentially, Ivanković et al. (2021) argue we need a more rhetorical understanding of actionable metrics. Although neither set of authors refer to “rhetoric,” both are closely circling the topic; the goal of actionable metrics is to *persuade* the audience toward action by showing how a potential individual action will impact an outbreak at a holistic level. To be effective, COVID-19 data dashboards must provide a way to look at data both holistically and locally, simultaneously, for an individual to make day-to-day decisions regarding their exposure to others. How, then, do COVID-19 data dashboards employ actionable metrics, what individual actions do these metrics potentially promote, and how are those individual actions tied to the holistic perspective of the outbreak?

To address these questions, first, I provide a brief history of the origin of disease maps, focusing on John Snow’s map of the 1854 cholera outbreak in London, exploring how the holistic-localist tension played out in early disease maps and what actionable metrics emerge from this tension. Next, I use the brief history as a basis for showing what visual rhetorical conventions have persisted from early disease maps into contemporary disease data visualizations and how these conventions are used to visualize actionable metrics. Then, I conduct a close rhetorical analysis of two popular COVID-19 data dashboards from my local perspective to understand what potential actions a viewer might be persuaded to take in my geographic context. I find that the data in both dashboards privilege a holistic view of data in a way that downplays risk, and although actionable metrics could be present, they lack sufficient contextual data to interpret them in one’s individual context, thus are not likely to be actionable. In conclusion, I suggest that for actionable metrics to be actionable, they must take advantage of the holistic-localist tension inherent in disease maps – to present data in a way that a user can view data from a local and holistic perspective simultaneously to understand how their past actions or potential future actions could impact the overall course of an outbreak.

Holistic-localist tension and actionable metrics in early disease maps and COVID-19 data dashboards

Disease maps emerged as a significant public health tool in the early 19th century, when a miasma theory of infection was the prevailing paradigm.¹ Advocates of the new germ theory used disease maps to illustrate the uneven distribution of cases (looking holistically), thus pinpointing one or more potential microbial sources of infection (looking locally). John Snow is acknowledged as the first to successfully use a disease map to identify the source of an outbreak. Importantly, he not only

pinpointed the potential cause of the 1854 cholera epidemic in London, the Broad Street water pump, but he also made a compelling case for germ theory as an explanation of disease transmission (Welhausen, 2015a).

A crucial difference between uses of COVID-19 data visualizations and Snow's map is that it was not necessary for Snow to engage with the local community to enact an effective public health intervention, as would be the case today to advocate for vaccines and mask use. Undoubtedly, those who relied on this pump were certainly confused and even angered by the removal (or at the very least, thirsty), but it was Snow's knowledge, as mediated by the map, and the social privilege he enjoyed as Queen Victoria's anesthesiologist that enabled him to single-handedly (arguably) curb the outbreak (cf. Tufte, 1997). The primary audience was Snow, himself, and eventually his colleagues in public health. The case of John Snow epitomizes Miles Kimball's argument that "information graphics inherently privilege the viewpoints of the powerful; they are used from a position of strength, representing corporations, agencies, classes, and institutions trying to forward agendas, prove points, and solve problems" (Kimball, 2006, p. 379).

Like its predecessor, the rhetorical purpose of COVID-19 data dashboards is to help the viewer identify geographic locations of high disease transmission by mapping cases, hospitalizations, and/or deaths (looking holistically) and make decisions regarding intervention or behavior (localist thinking). However, the audience of COVID-19 data dashboards is dramatically broader and more heterogeneous than disease maps. Early disease maps, such as John Snow's map of the 1854 London cholera outbreak, had a primary audience of one, the author, and a singular actionable metric, the physical location of the source of the outbreak. By contrast, the "About" page at the now closed Johns Hopkins Coronavirus Resource Center (JHU CRC) states, "We collect and analyze the best data available on cases, deaths, tests, hospitalizations, and vaccines to help the public, policymakers, and healthcare professionals worldwide respond to the pandemic" ("About Us", n.d.). Both audience and potential actionable metrics are intentionally broad for JHU CRC. Policymakers and healthcare professionals can use the dashboards to understand transmission of disease across large geographic or demographic areas to develop effective intervention strategies. Nonexpert potential patients and healthcare providers may use dashboards to understand their likelihood of infection at any given time. COVID-19 data dashboards are pulled between the purpose of facilitating a holistic understanding of the movement of a pandemic and the individual-level risk of illness as well as contribution of individual action to the larger pandemic. A single user, to meet all their information needs, would theoretically need to move back-and-forth from holistic to localist views.

The holistic-localist tension reflects what Barton and Barton (1993) would call the dual panoptic modes of visual surveillance: the synoptic and analytic. Based on Foucault's metaphor of the panopticon for a disciplinary society, the synoptic mode captures the entirety of the surveilled into view of a single, unseen viewer. Barton and Barton identify the map as the paradigmatic example of the synoptic mode of power. The analytic mode involves the quantification of individuals, or numbers. In the context of the Panopticon as a technology of prisoner surveillance, the analytic mode is "necessary to analyze, impose discipline, and correct behavior at the level of the *individual* inmate" (Barton & Barton, 1993, p. 141, emphasis in original). Barton and Barton extend Foucault's metaphor to develop their metaphor of data visualizations as power inscriptions. Their metaphor has provided a productive way to deconstruct unethical applications of data visualizations in our surveillance society. However, applied to disease maps, the metaphor of power fits well with the metaphors of "control" and "containment" that are widely used in public health – though these metaphors were quickly rendered moot with the speed at which COVID-19 spread globally. Nonetheless, disease surveillance by way of data visualization can be a powerful tool for public health. With regard to COVID-19 data dashboards, public health officials, epidemiologists in particular, might use dashboards primarily for the synoptic capability or to surveil large geographic and/or demographic areas to design effective intervention strategies. Nonexperts may use dashboards primarily for the analytic mode: to understand what they, or a loved one, might do to mitigate their risk of infection or potential to infect others at a given time or place.

It should be of great interest to rhetoricians that the disease map shifts toward nonexpert audiences for purposes of not necessarily surveilling, per se, but determining appropriate action during COVID-19 and later times. The primary audience for data dashboards has exponentially expanded from a single individual (in the case of John Snow) to a multidisciplinary heterogeneous audience of wide-ranging subject area expertise. Likewise, the stated purpose of data dashboards has equally expanded from determining a singular actionable metric (the pinpointed source of an outbreak) to a wide range of potential actionable metrics that would inform both large-scale, innovative interventions and shifting day-to-day decisions by individuals regarding their level of exposure.

The impact of visual conventions on actionable metrics and shifting audiences' risk perception

Conventions of Snow's map survive well after a germ theory of disease prevails, and principles derived from his original map of the cholera outbreak continue to inform the field of epidemiology today. As argued in past research, the convention of an aerial viewpoint, now coupled with a convention of warm and cool color use, can shift an audience's perception of risk, thus potentially shifting their actions to more or less risk-averse. Visual conventions, in this way, can signal actionable metrics to a viewer; therefore, it is important when analyzing visuals to consider conventions along with actionable metrics and the modes of power they employ.

Similar to Snow's cholera map, Welhausen (2015a) argues that early yellow fever maps invoked the synoptic mode of power over diseased space by employing conventions of the aerial view and shading to indicate density of the outbreak. She argues that these visual conventions helped build the principle of containment, which serves as the guiding metaphor of epidemiology today. As print and eventually digital media evolve, strategic use of color is added to the conventions of a disease map. As Welhausen (2015b) found in contemporary Ebola maps, the use of warm and cool colors can increase or decrease perception of risk, respectively, in part, due to color's inherent emotional appeal (Kostelnick, 2016). The emotional appeal of color in digital design, specifically, has the potential to increase a viewer's proximity to data, thus increasing their perception of individual risk (Kostelnick, 2016). However, when coupled with depictions of large geographic areas, as commonly done in disease maps, it invites an understanding of risk at a collectivist level, decreasing the viewer's proximity to data (Welhausen, 2015b). Although color included on these maps is helpful for epidemiologists who primarily look at the holistic level of disease transmission, it potentially minimizes the impact of an epidemic on an individual looking primarily at the localist level.

The convention to favor a collectivist understanding of disease risk is problematic in the context of COVID-19 data dashboards, as they were immediately public-facing and positioned to facilitate decision-making at several expert and nonexpert levels. Of course, the principle of containment that is facilitated by disease maps became a moot point in the COVID-19 pandemic as it quickly spread all over the globe. At that point, the refrain "flatten the curve" became the driving metaphor of the outbreak, and an individualist perspective of risk overshadowed the collectivist. The call-to-action, flatten the curve, accompanied instructions for individuals to limit exposure, increase quality and frequency of handwashing, and eventually don masks and seek vaccination. FLC demands buy-in from individuals, and as the pandemic droned on, many were left making hard decisions about returning to prepandemic activities and using masks based on their individual circumstances including competing risks and needs (Popli, 2022).

Data visualizations can facilitate individuals' decision-making regarding personal risk of disease and promote positive public health behaviors. For instance, Valdiserri and Sullivan (2018) show how data visualizations mobilize multiple audiences against AIDS/HIV. Additionally, Welhausen (2017) shows how disease tracking apps can enable decision-making on an individual level because they,

provide more individually-focused, location-specific information that aligns with the ways that non-experts perceive risk—that is, in terms of how the risk might personally affect them and/or their families. In this way,

digital disease-tracking apps facilitate a DIY risk assessment ethic in which people potentially affected by a particular risk choose what information they want to collect about the risk and consequently formulate their own risk assessment, which they can then use to engage in specific actions designed to mitigate the risk. (p. 60)

Data visualizations were evoked as decision-making tools with the first utterance of FLC. Functioning as both a conceptual metaphor and reference to actual case rates over time, the message oversimplifies data in ways that could impact visual risk literacy and deliberation (Amidon et al., 2021).

At the same time, the message is potentially powerful. Huiling Ding might liken calls to FLC as an effort to achieve a tipping point or moments of critical transitions that “jolt an epidemic out of equilibrium” (Ding, 2014, p. 39). Under the mantra FLC, COVID-19 data visualizations are framed around potential tipping points, where the number of new cases visually tips in a downward direction. For example, in the early days of the pandemic, when the primary control strategy was travel restrictions and self-isolation, the war-room-like data dashboard created by Johns Hopkins University visualized potential containment areas on interactive global maps. As the pandemic progressed to virtually all areas of the globe and containment became irrelevant, data visualizations incorporated new potential tipping points such as testing and testing sites, hospital resources, racial disparities, mask use, and later, vaccination rates – all potential actionable metrics that could help a viewer determine their localized and individualized level of risk at a given time.

To understand how COVID-19 data dashboards employ actionable metrics, what individual actions the metrics potentially promote, and how the individual actions are tied to the holistic perspective of the outbreak, I conducted a rhetorical analysis of two widely accessible COVID-19 data dashboards with different overall esthetics: the Johns Hopkins University Coronavirus Resource Center (JHU) and the Institute for Health Metrics and Evaluation (IHME). In an early quantitative assessment of COVID-19 data dashboards, these were the second and fifth-highest scoring data dashboards, respectively (Pietz et al., 2020). I collected screenshots of many visuals from each dashboard in the weeks of February 8 and 15, 2021, focusing on visualizations of my home county at the time: Allegany County, Maryland. In some instances, I was able to look at county-level data, and in some cases only state-level data. In what follows, I provide a brief overview of my geographical context and follow with an analysis of visuals collected from JHU and IHME.

Geographic context: allegany county, maryland

Allegany County is a rural area of Western Maryland, situated between rural West Virginia and Pennsylvania. Many Eastern Woodlands tribes inhabited this area, including the Monongahela, Iroquois (Haudenosaunee), Susquehannock, Shawnee, and Delaware (Lenni Lenape). Some colleagues live in neighboring states, and many of us choose to seek healthcare in places like Keyser or Morgantown, WV or Bedford, PA. In fact, the close proximity of states has caused some confusion with vaccine distribution for some who live in a neighboring state and attempt to seek an appointment in Maryland. Culturally, politically, demographically, and economically, Allegany County is much more similar to rural WV or PA than to the Baltimore/DC area or “down state,” the eastern- and southern-most bay area of Maryland. Although our geographic area is unique, it is worth pointing out just one instance that state-level data is relatively useless for understanding the status of one’s local community. There are certainly many more similar examples across the country.

Modes of power, holistic-localist tension, and their impact on actionable metrics in the Institute for Health Metrics and Evaluation (IHME) and Johns Hopkins University Coronavirus Resource Center (JHU CRC) data dashboards

The holistic-localist tension manifests differently in the Institute for Health Metrics and Evaluation (IHME) and Johns Hopkins University Coronavirus Resource Center (JHU CRC) data dashboards. IHME takes a more holistic approach with few discernable actionable metrics; JHU CRC, however,

provides more localist-based actionable metrics, but they are nearly severed from the holistic perspective of the outbreak and, thus, not actionable. For purposes of analysis, it is helpful to identify where modes of power are employed, identify the dimension of power, and then describe the connection to any actionable metric, if present. In this section, I first provide a broad description of each dashboard, highlighting the modes of power that each one employs. Then, I discuss what the modes of power employed by each mode reveal about the holistic-localist tension inherent in each dashboard and the impact on actionable metrics, if present. Next, I shift to how the visual conventions employed by each dashboard impact the actionable metrics. In the final section of my analysis, I review a unique feature of the JHU dashboard that does not have a comparable element on the IHME site: an autogenerated county-level infographic. Here, I consider the actionable metrics that are visualized on the infographic along with modes of power and visual conventions – with a little external research, revealing areas where a viewer may be potentially misled.

The Institute for Health Metrics and Evaluation (IHME)

IHME relies almost exclusively on the synoptic dimension in its data visualizations by privileging line graphs showing change over time alongside projections. On the home page of the IHME dashboard, the user can select a state from a drop-down menu for state-level data consisting of total deaths, daily deaths, hospital resource use, infections and testing, mask use, and social distancing. Line graphs show real data from August 2020 to early February, approximately 1 week before the date of viewing, and projections up to the start of June 2021. The portion of the line graph built on real data is solid, while projection data lines are dashed. A vertical dashed line marks the view date on the line graphs. The X-axis shows time by months and a vertical axis of count in intervals of 500. The projected data shows four different possibilities: “worse,” “rapid variant spread,” “current projection,” and “universal masks.” Each projection is a filter that can be turned on or off to be viewed singularly or in tandem. The user can hover their mouse over a future point of the projection to reveal a pop-up with exact counts of each projection at that specific point – perhaps the only manifestation of the analytic surveillance mode. Each projection is marked with an asterisk meaning “projection.” The counts are provided up to the hundredth decimal, with a range in italics and parenthesis afterward. Presumably, the range is the margin of error, again down to the hundredth decimal. [Figure 1](#) is a screenshot of the projection for total deaths in Maryland, taken February 15, 2021.

IHME provides three ways to look at each data point: the trend, comparison, and map. The user can flip between these views by clicking the corresponding icon in the top right corner, as depicted in [Figure 1](#). The “trend” line graph for daily deaths, consistent with other line graphs on the site, shows observed daily deaths in a smoothed solid line with four different scenarios of future projections in a dashed line. A shaded area shows, presumably, the potential range or margin of error. The “comparison” line graph enables the user to compare daily deaths in Maryland to daily deaths in the US or other individual states. The “map” option provides a map of the U.S. showing daily deaths in each state per 100,000 population. States shaded in the lightest pink have 0–0.18 deaths per 100,000. The darkest red states have 3.0–4.9 deaths per 100,000. The user can click a play button to animate the map to move quickly through data from August 2020 to projections up to June 2021. [Figure 2](#), retrieved February 15, 2021, shows the animation paused at the projection for April 14, 2021.

Each line graph on the IHME dashboard, with some undulations, indicates that Maryland has achieved a tipping point, with the curve moving in the downward direction. The options to compare to other states indicate that tipping points are achieved at different times for each state. Comparing states may be useful for national public health officials who need choose where to focus efforts, geographically. For public audiences, comparing states communicates a successful containment campaign to “flatten the curve” as the user can see curves for Maryland as well as other states tip in the downward direction. In the animated map, the user can watch states literally flare up in dark red from earlier months then progressively get lighter – visually indicating a subsiding level of risk at the time.

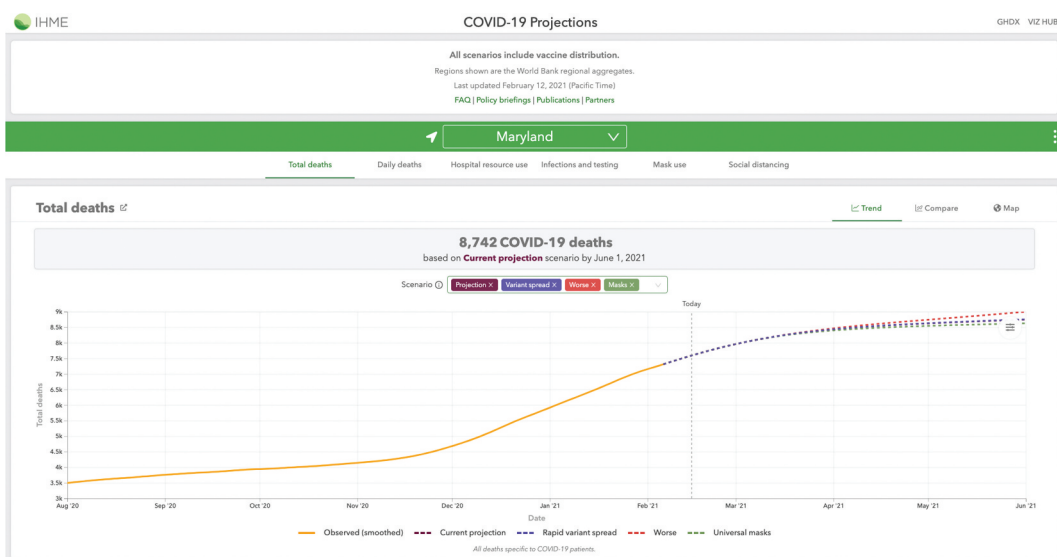


Figure 1. IHME total COVID-19 death projection for Maryland, retrieved February 15, 2021.

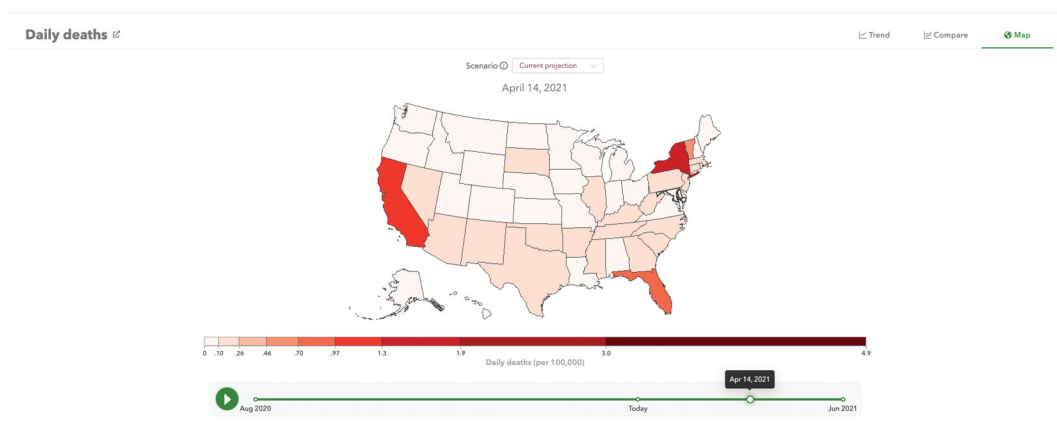


Figure 2. IHME daily COVID-19 death projection for Maryland, retrieved February 15, 2021.

Johns Hopkins University Coronavirus Resource Center (JHU CRC)

Johns Hopkins University Coronavirus Resource Center (Figure 3) provides a more panoptic view (using both synoptic and analytic modes in tandem) of the COVID-19 pandemic, unlike IHME. Both modes are present on the starting page; at the center lies a dark gray global map with red bubbles in sizes proportionate to the number of cases in the region, bar/line graph hybrids show weekly cases (red), weekly deaths (white), and weekly vaccine doses administered (green). The same colors are used across the top of the graph to show total counts of each category and 28-day counts, employing the analytic mode. The left pane gives a list of countries with 28 days and total case and death counts in white and red, respectively. The viewer can zoom in to county-level in the United States, and top 50 counties are ranked on the left with total confirmed cases in red, ranked highest to lowest, and top 20 counties in deaths are listed on the right. When I zoom in to my home county of Allegany, Maryland, I can see Allegany and surrounding counties in varying shades of red, presumably based on the total case counts.

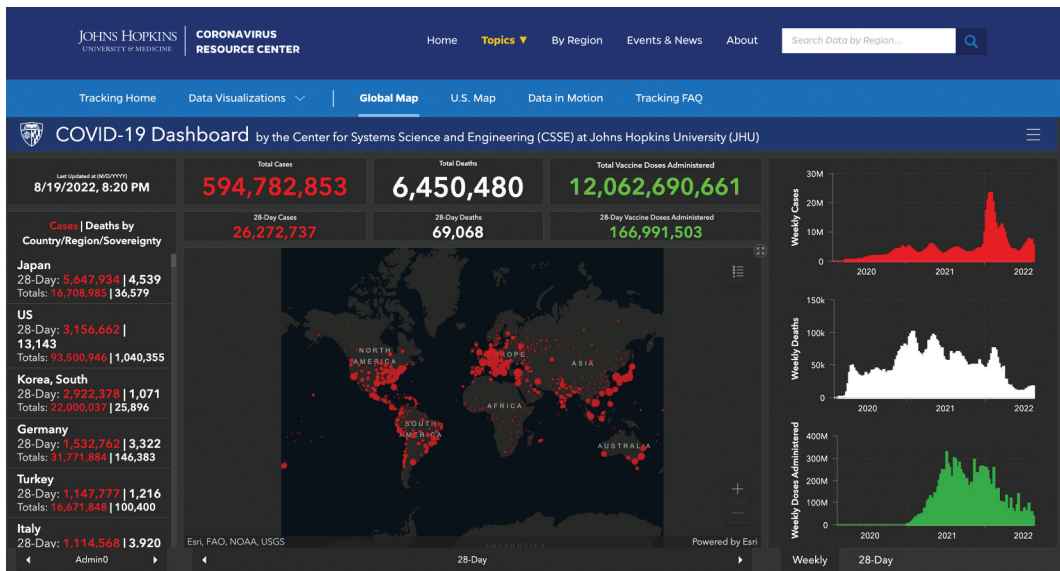


Figure 3. Johns Hopkins University Coronavirus Research Center home page, retrieved August 19, 2022.

Other than total vaccination count, JHU does not include actionable metrics on the main landing page. However, there are some useful figures accompanying articles elsewhere on the site, but these are still from a synoptic, holistic level and not from the perspective of an individual's potential action. Most charts visualizing interventions only look at national levels, and a few that look at state-level data. For example, the page, "Critical Trends" includes a subpage, "Impact of opening and closing decisions by state" (refer to Figure 4). Once Maryland is chosen from the drop-down menu, a line graph appears depicting new daily cases (Y-axis) over time (X-axis) with superimposed opening (green vertical dashed line) and closing policies (red vertical dashed lines).

The page includes two paragraphs explaining how to use the visualization. As instructed, the viewer can hover over a point on the timeline to see what policy was in place and how many new cases were reported that day. The primary purpose of this graph is to assess state leaders' role in controlling the pandemic. For Maryland, it would appear that Governor Hogan has done a reasonably good job at closing when cases were on the rise and opening when cases were on the decline. What is not immediately apparent is mask mandates; if relevant to the state, these policies would be visualized by a gray dot, then described under the graph once clicked. The viewer would be required to click through all gray dots to see which one represented mask mandates. Of course, mask "mandates" are not necessarily an indicator of whether masks are widely used. Likewise, opening and closing decisions do not necessarily correlate to *individual compliance* to opening and closing mandates. Thus, this visualization is limited in illustrating the impact of individual action in flattening the curve.

Modes of power, Holistic-Localist tension, and impact on actionable metrics

Both IHME and JHU are positioned primarily from a synoptic mode of power – privileging the holistic view of data – with few actionable metrics positioned from an analytic (or individual) mode of power. Actionable metrics are limited; IHME makes a subtle suggestion of mask use – with what is framed as a slight improvement in deaths with universal masks. But how much control does an individual really have over "universal" mask use? How would an individual know just how universal mask use is in their current area? One avenue for future research may be to explore how reporting data on self-reported mask use over time may impact an individual's likelihood to wear a mask or not. It might be more

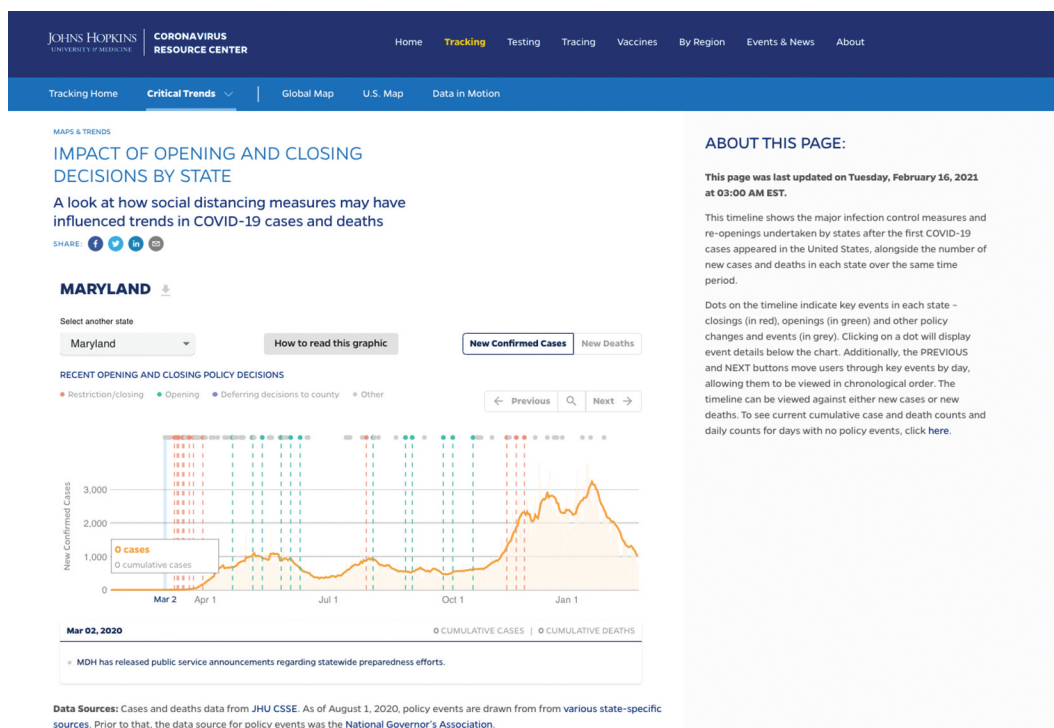


Figure 4. “Impact of opening and closing decisions by state” taken from JHU Coronavirus Resource Center, February 16, 2021.

motivating for an individual to wear a mask if they can see, through data visualization, a negative correlation between case counts and self-reported mask use in their local community.

Similarly, JHU makes a subtle suggestion of vaccination by tracking dose counts. But again, an individual will be more persuaded to seek a vaccine if these counts are overlayed case or death counts. If I regard vaccination as normalized in my community and/or I regard vaccination as negatively correlated to case and death counts, I may be more likely to seek the vaccine. Admittedly, February 2021 may have been too soon to show the correlation, as most of the available vaccines at that time required multiple doses and the time to reach full efficacy.

Modes of power, visual conventions, and impact on actionable metrics

JHU CRC positions the user from an aerial perspective, as is common with disease maps. Similarly to IHME, it offers a way to directly compare countries, states, counties, and major cities. The use of bubbles rather than solid shading of countries or states may remind a user that political borders are irrelevant to a pandemic. It may also have the effect of communicating the outbreak in terms of “hot zones” that radiate evenly outward from a singular source. After zooming in to a specific county, the bubbles are replaced with even shading following county borders; even shading that follows county borders is likely an artifact of data being contributed by local county health departments.

I have mentioned some conventional uses of color in the previous section in both IHME and JHU CRC – red is generally used to indicate something bad, a high risk, or closing (i.e., the closing policies in Figure 4); green is generally used to indicate something is good, low or no risk, or opening (i.e., opening policies in Figure 4). JHU CRC uses shades of red that are used to indicate case counts at the county level, and proportional, clustered red bubbles to illustrate levels of outbreak in different geographic areas at the global level. As one zooms out, the bubbles cluster, showing a more condensed

hot zone. The combination of color and bubbles invites a holistic perspective of risk at a regional or global level – as I look globally, the cluster of bubbles composed by the central East Coast/Appalachia region visually appear to impact me in Maryland, as we collectively make a larger hot zone than Maryland alone.

IHME’s prioritization of line graphs breaks from the convention of using aerial maps. Use of line graphs instead of aerial maps minimizes the individual’s risk by removing the state from its regional context and erasing the more granular county data. At the same time, this enables the viewer to look at both real data and projection data simultaneously in a way JHU CRC does not allow. A user might look at the total COVID-19 death graph (Figure 1) and be reasonably comforted that around May the predicted slope of all four projections looks close to 0, a plateau. However, closely looking at the projection counts, there is a difference of nearly 200 deaths for Maryland between the best-case scenario, “universal masks,” and worst-case scenario, “worse.” It is not clear how “worse” is being defined by modelers and, if it is, indeed, worst case scenario. Shortening the X-axis or lengthening the Y-axis could provide a remedy, as it would visually increase the slope to better visualize that it is not actually plateauing, and there is a great difference between the best and worst projections.

Daily death count, shown in Figure 5, shows a different answer than in Figure 1. All four projections steadily decrease, but none of them reach zero by June. A user might look at this figure and, reasonably, feel hopeful for socially distanced outdoor gatherings the following summer, but would not hang up their mask just yet, especially when we consider that the projection for “worse” is *triple* the projection for “masks” on June 1 (approximately 9 and 3, respectively).

One of the premises of the FLC argument is that if the curve is significantly steep, hospitals and resources would be strained. A civic-minded individual who is generally healthy might consider this metric as the best for determining when one can lower protective measures. For this reason, I am glad to see that IHME includes a hospital resource use a graph, another break from convention of the disease map. However, this graph is misleading; presumably, the model does not account for regions that may already be underserved medically. As shown in Figure 6, this graph does not delineate what is based on real or projected data. Because the entire graph is a dashed line, one could assume it is based entirely on projected data, but it is unclear. The X-axis remains the same for all charts, and the Y-axis in Figure 6 is “resource count” in intervals of 500. It is unclear what resources are counted, but one would presume beds and ventilators, as total counts of “all beds available” and “ICU beds available,” are indicated with purple and green horizontal lines, respectively. Dashed lines show “all beds needed,” “ICU beds needed,” and “invasive ventilators needed.” In light, italicized font under the key is the important variable that “All resources specific to COVID-19 patients.” A user might look at this graph and determine that the beginning of March is a safe time to lower one’s caution, as that is the projected

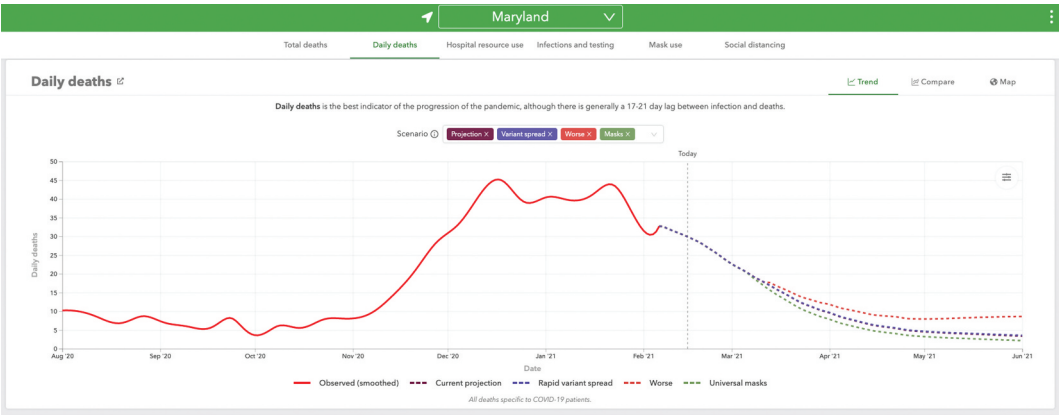


Figure 5. IHME daily death projections for Maryland.

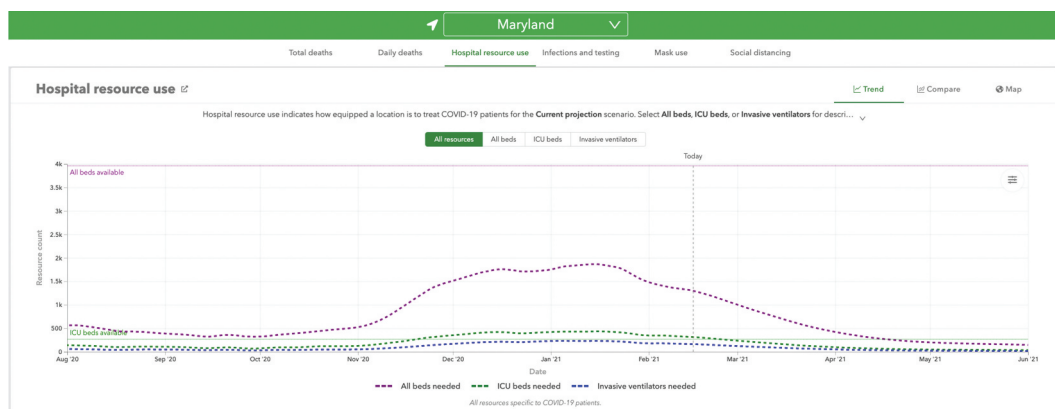


Figure 6. IHME hospital resource use for Maryland.

date that “ICU beds needed” will dip below “ICU beds available.” However, this graph does not come with the caution that ICU beds could be overwhelmed by nonCOVID-19 related cases: a surge in influenza cases, a major vehicle pile-up, a terrorist event, a natural disaster, etc. Additionally, because this graph displays state-wide resource counts, it also does not account for regions that could already have understaffed and under-resourced hospitals or regions that experience local spikes in cases due to some unique local variable such as attitudes toward social distancing, local cultural or religious events, outbreaks in long-term care facilities, etc. In short, there may be a bed available to me *somewhere* in the state, but that bed may or may not be at a local hospital that accepts my insurance. Again, as stated earlier, using line graphs to remove state-level data from its regional context and erase more granular county-level data seems to do more disservice than the convention of aerial maps with large swaths of color to indicate a level of risk. It oversimplifies data to the point that it potentially misleads a viewer to a risky decision.

IHME includes “mask use” and “social distancing” as two potential actionable metrics, but the effectiveness of each is minimized by the graphs provided. The viewer is assumed to understand that both strategies work, but little evidence is given to support these measures. At the same time, no contextual data other than hospital resource use – presumably beds, not staff, ventilators, or PPEs – is provided to help an individual to develop their personalized risk assessment. A viewer, again, would be on their own to understand the overall risk level of their area in terms of its overall health, aging populations, populations of color, and populations with comorbidities. IHME downplays the risks of not complying with social distancing or mask use by visually minimizing its own projections and not providing sufficient contextual information for a viewer to develop a personal risk perception. IHME appears to assume viewers will simply comply with recommendations or not, rather than developing a nuanced risk perception based on their specific context. Tracking impact of COVID-19, for IHME, involves total deaths, daily deaths, infections and testing, and hospital resource use. All these metrics are positions from a typical expert-audience risk perception – that is, quantitatively at no smaller than the county level (Welhausen, 2015b).

JHU and IHME rely on conventional color schemes to communicate level of risk. JHU’s use of bubbles overlayed on a spatial map, in tandem with color, does more to communicate health risk than IHME’s emphasis on line graphs with state-level data. IHME, to their credit, emphasizes social (i.e., hospital and resource use) and behavioral (i.e., masks) factors, and their importance, which are minimized, by presenting them in line graphs and thus removing them from regionally contextual data (i.e., surrounding states and county-level information).

Thinking in terms of actionable metrics, one might believe it sufficient to include data on a metric that is in the control of the individual (in this case, mask use and vaccinations) as both IHME and JHU have done. IHME even goes a step further by providing future projections based on behavioral

interventions (universal mask use). However, considering actionable metrics in tandem with modes of power and visual conventions, we can see a single metric in control of the individual is not sufficient. JHU's simple inclusion of vaccination rates at the analytic level (i.e., a running count of total doses) without overlaying it with the more nuanced tracking of cases and deaths on an aerial map (thus employing the synoptic mode of power) does not show the impact of an individual's choice to vaccinate on the community or on the holistic trajectory of the pandemic. IHME's break from convention by employing line graphs in lieu of an aerial map, lack of explanation of their "universal masks" projection, and lack of reporting self-reported mask use over time minimizes the potential impact of that simple behavioral intervention on the course of the pandemic.

Actionable metrics on JHU CRC county-level infographics

JHU has a unique feature on its dashboard that is worthy of its own section for analysis: an auto-generated county-level infographic found if a user clicks on the county from the main map ([Figure 7](#)). The design of the graphic is consistent with the design of the JHU dashboard – dark gray background, white text including the JHU branding, and a palette of warm and cool colors. Starting from the left side under the title, the first vertical bar chart, "Healthcare Facts," shows the proportion of the adult population covered by different types of health insurance or no insurance; each bar is further divided by age. It is easy to read that approximately 5% of the population is uninsured, none of which is 65 or older. No mention of the proportion of uninsured minors appears anywhere on the graphic. The center of the graphic contains a cluster of three bar graphs collectively titled "Population Race, Ethnicity and Age." The central graph, ethnicity, has a somewhat stretched Y-axis, perhaps so the small proportion of "Hispanic or Latino" can be viewable. It is easily seen that Allegany County is majority non-Hispanic white; according to the U.S. Census, the county is disproportionately non-Hispanic white compared to the national average: 86.7% in Allegany County and 60.1% in the United States (U.S. Census Bureau, 2021).

The age graph is curiously divided. The key primarily contains groups spanning 9 years except for under 15, over 75, and age group 35–64, spanning 29 years. The bar reflects this peculiarity, with the group 35–64 being largest in a gold color. Knowing that age group 65 and

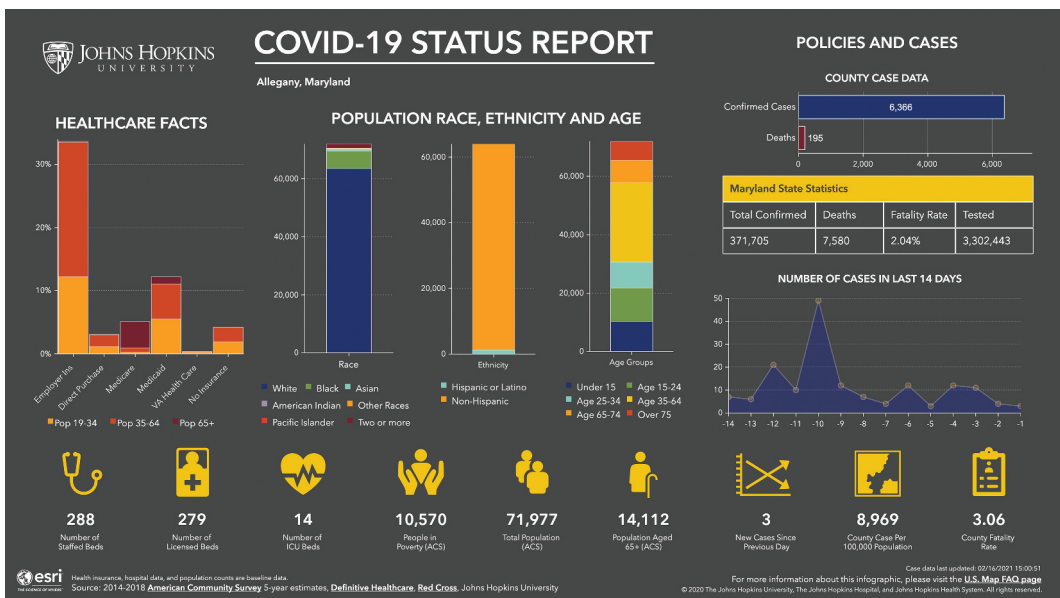


Figure 7. Johns Hopkins University COVID-19 infographic for Allegany County, Maryland, retrieved February 16, 2021.

older is most likely to experience severe illness, hospitalization, and death, and knowing that children, at the time this infographic was generated, are less likely to be severely impacted, the use of color appropriately reflects risk: the youngest group in dark blue and the oldest group in dark red.

Three graphs clustered vertically on the far right are titled “Policies and Cases,” though they contain no information about policies. The top graph of this group, “County Case Data,” provides the raw number of confirmed cases and deaths. Color is not used strategically, as “Confirmed Cases” are in dark blue, associated with low risk. The middle table of this cluster, titled “Maryland State Statistics,” also does not use color appropriately – a warm yellow-gold is used for the title, but the raw numbers provided do not have enough additional information to interpret whether the county is worse, the same, or better than the state on average (with the possible exception of fatality rate, discussed in the next paragraph). The final graph in this cluster, “Number of Cases in Last 14 Days,” presumably *new* cases, shows a spike 10 days previous but appears to be lower since, and either plateauing or declining at the day the infographic was taken. The graph is shaded blue beneath the line, but it is unclear if this color (associated with little to no risk) should be read meaningfully given the unstable numbers of the previous at least 10 days.

Nine gold-colored icons across the bottom of the canvas accompany numbers of staffed beds, licensed beds, ICU beds, people in poverty, total population, population aged 65+, new cases since previous day, county case per 100,000 population, and county fatality rate. While five of the nine icons accompanying the statistics consist of a humanized representation, two of the icons representing COVID-19 case rates do not (“New Cases Since Previous Day” and “County Case Per 100,000 Population”) and a third (“County Fatality Rate”) contains a human figure so small it is nearly unnoticeable. Because these are statistics that reflect actual people sick, hospitalized, or dead, it is ethically imperative that these are humanized (Dragga & Voss, 2001). Additionally, the gold color of all these icons reads neutral, which is another missed opportunity to use color strategically to communicate risk level according to different contextual factors. With further investigation using outside sources, these data points are anything but neutral:

- (1) Over 20% of the population is 65 and over, higher than the national rate of 16.5% (US Census Bureau, 2021).
- (2) Almost 15% of Allegany County is in poverty, higher than the 2020 national poverty rate of 11.4% (US Census Bureau, 2021).
- (3) There are four staffed hospital beds per 1000—better than national average of 2.4 per thousand, and even better than state average of 1.8 per thousand (Masson, 2021). However, according to the website of the only hospital in the county, it serves Allegany, Garrett and surrounding counties in WV and PA (UPMC Western Maryland, n.d.). Combining the population of Allegany, Garrett, Mineral, and Somerset, the rate is 1.5 beds per thousand, 37.5% lower than the national average, and 16.67% lower than the state average (Masson, 2021; US Census Bureau, 2021).
- (4) Nearly 14% of Allegany County residents with a disability are under age 65, compared to the national rate of 8.7% (US Census Bureau, 2021).

In sum, the county is older, more impoverished, and more disabled than the country as a whole, and has fewer staffed hospital beds than the state and national averages. If given this extra contextual information, I would decide to take extra precaution in terms of masking and social distancing, especially if visiting areas where I might interact with those most vulnerable to COVID-19.

The county fatality rate is higher than the state fatality rate, which can be gleaned from comparing the icon at the bottom right to the “Maryland State Statistics” (refer to Figure 7). Even simply juxtaposing these two points would much more clearly send a message that Allegany County is higher risk than elsewhere in the state.² Although these two points can be directly compared, the visual distance and difference in formatting discourages the viewer from doing so.

Overall, the JHU CRC fails at providing sufficient contextual information for an individual to evaluate their personal risk or their potential impact on the pandemic on a holistic level. To their credit, the infographic generated by the JHU CRC provides raw statistics for several important social factors (people in poverty, race, ethnicity, and age), and some pandemic resources (staffed beds, licensed beds, ICU beds). However, these raw statistics are not useful without additional information to interpret them, as I have described previously. Furthermore, without overlaying this information on a more synoptic view at the county, state, national, or global level, an individual does not see how their behavior (e.g., self-isolation, mask use, vaccination) impacts their community or the pandemic as a whole. With additional information, it is reasonable to conclude that Allegany County is at high risk of disastrous pandemic fallout, and, thus, its residents should act with greater precaution and adopt positive public health behaviors.

Conclusion

The Institute of Health Metrics and Evaluation and Johns Hopkins University Coronavirus Resource Center both rely primarily on a synoptic mode of power in communicating data surrounding the COVID-19 pandemic. IHME privileges a view of case numbers as they change over time; JHU privileges a view of case numbers as they are distributed geographically. Relying on this mode of surveillance invites a collectivist understanding of risk, the perspective generally favored by public health professionals, not nonexperts. JHU enables the viewer to drill down to the county-level for additional information in the analytic mode of surveillance. However, this information, when contextualized with information from other sources, is misleading in the direction of minimizing individual and community-level risk. Neither platform provides sufficiently contextualized information regarding an individual's level of risk based on local or social factors. Additionally, little, if any, information is provided about specific interventional behaviors (social-distancing and mask use) and the impact on the trajectory of the pandemic overall. IHME and JHU fall tragically short in potentially empowering nonexperts to assess their individual risk and adopt positive interventional behavior.

Coronavirus data displays, like the Johns Hopkins Coronavirus Resource Center and Institute of Health Metrics and Evaluation COVID-19 models analyzed here, should take advantage of multi-modal affordances to enable a viewer to develop an individualized risk assessment and promote positive public health behaviors. Actionable metrics, as promoted by Verhulsdonck and Shah (2021) and expanded by Ivanković et al. (2021), can be a valuable mechanism for persuading an individual toward actions to help FLC. However, as I hope my analysis has revealed, these metrics need to fluctuate between the holistic and individualist perspectives by employing synoptic and analytic modes of power. Considered use of the conventions of the disease map, specifically color and aerial maps, can help an individual viewer see both their individual level of risk at any given time and the potential impact of their decision on the pandemic as a whole, thus potentially persuading the individual toward that positive interventional behavior.

Although I selected JHU and IHME according to recognized usability, availability, and credibility, they are not the only dashboards that lacked sufficient persuasive capability. Pietz et al. (2020) found that all dashboards they considered “lack clarity as to how intervention measures affect the spread of the virus” (p. 394). Of several recommendations, they suggest that governments

incorporate more action-oriented data that communicate the timing and stringency of response measures taken in order to give citizens a clear understanding of why compliance to governmental intervention policies is needed. Consistently present these data, along with disease status data, in a single dashboard (or similar data report). (p. 400)

The effort it took to present COVID-19 data in a reasonably digestible way at the speed at which IHME and JHU were able to do should be applauded. Certainly, the decision to make public health data public in such a manner, with real-time updates, provided some reassurance to many in such a gravely uncertain time – even if the reassurance was not a positive one. Data dashboards

have the potential to be useful for the foreseeable future – even as (perhaps especially as) the disease of focus necessarily shifts. The call to action to FLC is applicable to any communicable disease – and to act, we need data presented in a valuable, actionable way. Data dashboards can fill that information need. As conventions are refined and audiences become more familiar with actionable metrics, they have great potential as public health interventions. Personally, I am excited to see what new data visualization innovations are spurred from the COVID-19 data-tracking years.

There are several limitations to this analysis. First, dashboards are living documents; they are regularly updated in data and design features; an analysis at a different point or focused on a different geographic location could point to different conclusions simply due to the change in the dashboard. Second, an analysis at a different point in time of a pandemic that has proven to resurge again and again would yield different results. Lastly, this study is a first step toward understanding how viewers might potentially act on the information. A usability study, perhaps using key elements from this study, is a critical next step.

Notes

1. Welhausen (2015a) provides a high-level summary of early medical cartography.
2. In an open faculty Q&A with an Allegany County public health official, hosted by the Office of the President at my institution, I inquired about this point. The official shared that this difference is primarily due to the overall age of legal residents, and the fact that cases are reported in county of permanent residence. For instance, if an elderly individual falls ill while residing in a communal living facility or with a relative in a different county, their case and potential death are reported in Allegany if that remains their legal residence.

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